

TECNOLOGÍAS DE LA INFORMACIÓN

Desarrollo de Sistemas Informáticos

4 créditos



Profesores:

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Titulaciones	Semestre
• <i>Tecnologías de la Información en Línea</i>	<i>Quinto</i>

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PARALELO: A

SEMESTRE: 5

ACTIVIDAD: DESARROLLO DEL BACKEND Y BASE DE DATOS (API REST)

PERIODO ABRIL 2026 - AGOSTO 2026



Actividades a desarrollar en la Unidad 4

Actividad # 8

1. INTRODUCCIÓN

El presente informe describe la implementación del backend del Sistema de Gestión de Incidentes Help Desk mediante una API REST desarrollada con Node.js, Express, Sequelize y SQLite. El sistema permite gestionar tickets mediante operaciones CRUD.

2. OBJETIVOS

Objetivo general:

Implementar un backend API REST para administrar incidentes técnicos.

Objetivos específicos:

- Configurar la base de datos SQLite.
- Implementar modelos Sequelize.
- Crear endpoints CRUD.
- Validar la API mediante Postman.
- Aplicar control de versiones Git.

3. TECNOLOGÍAS UTILIZADAS

Node.js: ejecución del servidor.

Express: creación de servicios REST.

Sequelize: comunicación con base de datos.

SQLite: almacenamiento relacional.

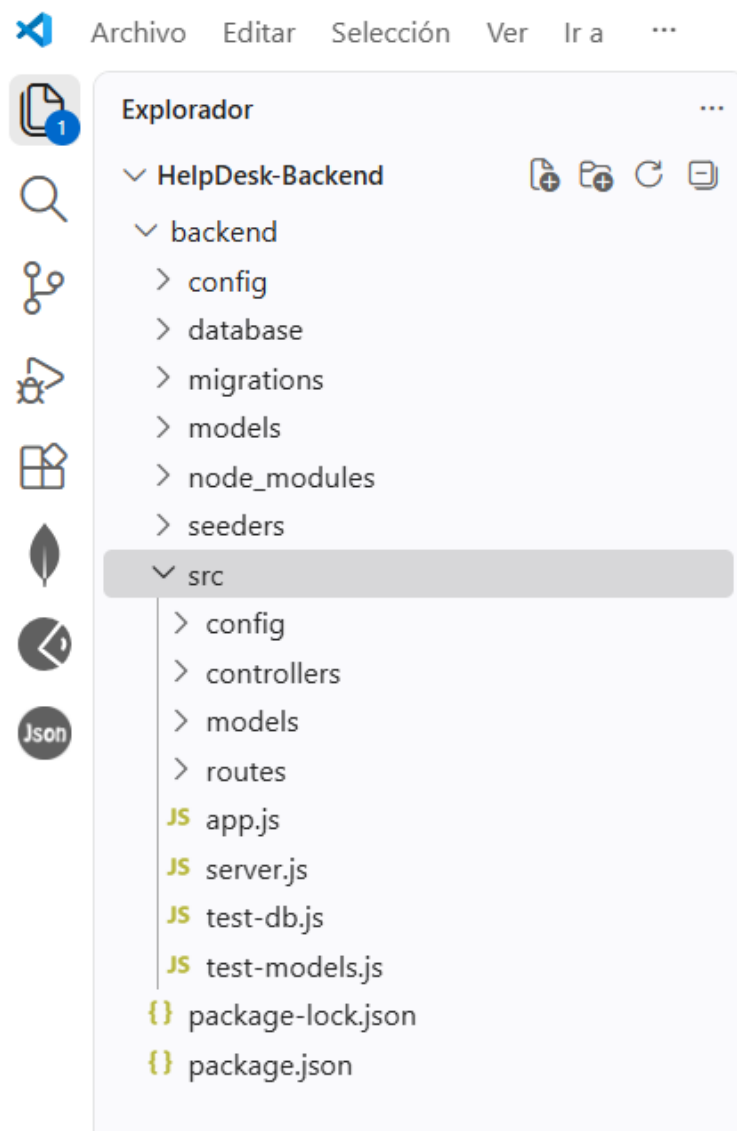
Postman: pruebas de endpoints.

Visual Studio Code: desarrollo.



4. ESTRUCTURA DEL PROYECTO

```
backend/  
├── src/  
│   ├── config/  
│   ├── controllers/  
│   ├── models/  
│   ├── routes/  
│   ├── app.js  
│   └── server.js  
├── database/  
└── package.json
```



CAPTURA 1: Estructura del proyecto en VS Code



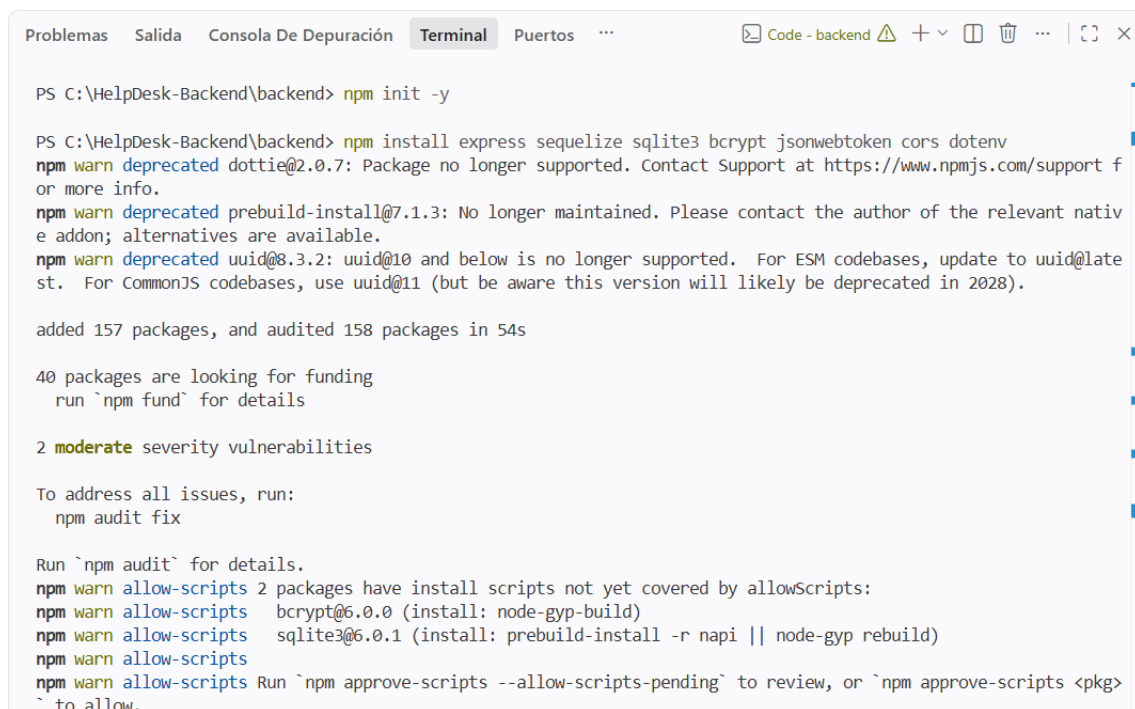
5. CONFIGURACIÓN DEL ENTORNO

Comandos utilizados:

```
node -v  
npm -v  
npm init -y
```

Instalación de dependencias:

```
npm install express sequelize sqlite3 bcrypt jsonwebtoken  
cors dotenv
```



```
Problemas Salida Consola De Depuración Terminal Puertos ... Code - backend + v [ ] [ ] ... [ ] [ ] X  
  
PS C:\HelpDesk-Backend\backend> npm init -y  
  
PS C:\HelpDesk-Backend\backend> npm install express sequelize sqlite3 bcrypt jsonwebtoken cors dotenv  
npm warn deprecated dottie@2.0.7: Package no longer supported. Contact Support at https://www.npmjs.com/support for more info.  
npm warn deprecated prebuild-install@7.1.3: No longer maintained. Please contact the author of the relevant native addon; alternatives are available.  
npm warn deprecated uuid@8.3.2: uuid@10 and below is no longer supported. For ESM codebases, update to uuid@latest. For CommonJS codebases, use uuid@11 (but be aware this version will likely be deprecated in 2028).  
  
added 157 packages, and audited 158 packages in 54s  
  
40 packages are looking for funding  
  run `npm fund` for details  
  
2 moderate severity vulnerabilities  
  
To address all issues, run:  
  npm audit fix  
  
Run `npm audit` for details.  
npm warn allow-scripts 2 packages have install scripts not yet covered by allowScripts:  
npm warn allow-scripts   bcrypt@6.0.0 (install: node-gyp-build)  
npm warn allow-scripts   sqlite3@6.0.1 (install: prebuild-install -r napi || node-gyp rebuild)  
npm warn allow-scripts Run `npm approve-scripts --allow-scripts-pending` to review, or `npm approve-scripts <pkg>` to allow.
```

CAPTURA 2: Instalación de dependencias

6. CONEXIÓN SQLITE – SEQUELIZE

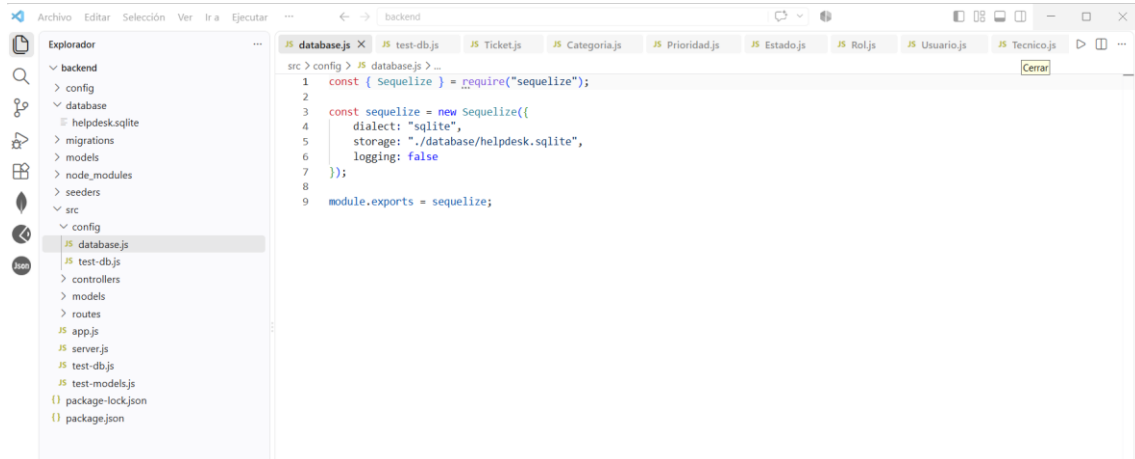
Archivo: src/config/database.js

```
const { Sequelize } = require('sequelize');  
  
const sequelize = new Sequelize({
```



```
dialect: 'sqlite',  
storage: './database/helpdesk.sqlite',  
logging: false  
});
```

```
module.exports = sequelize;
```

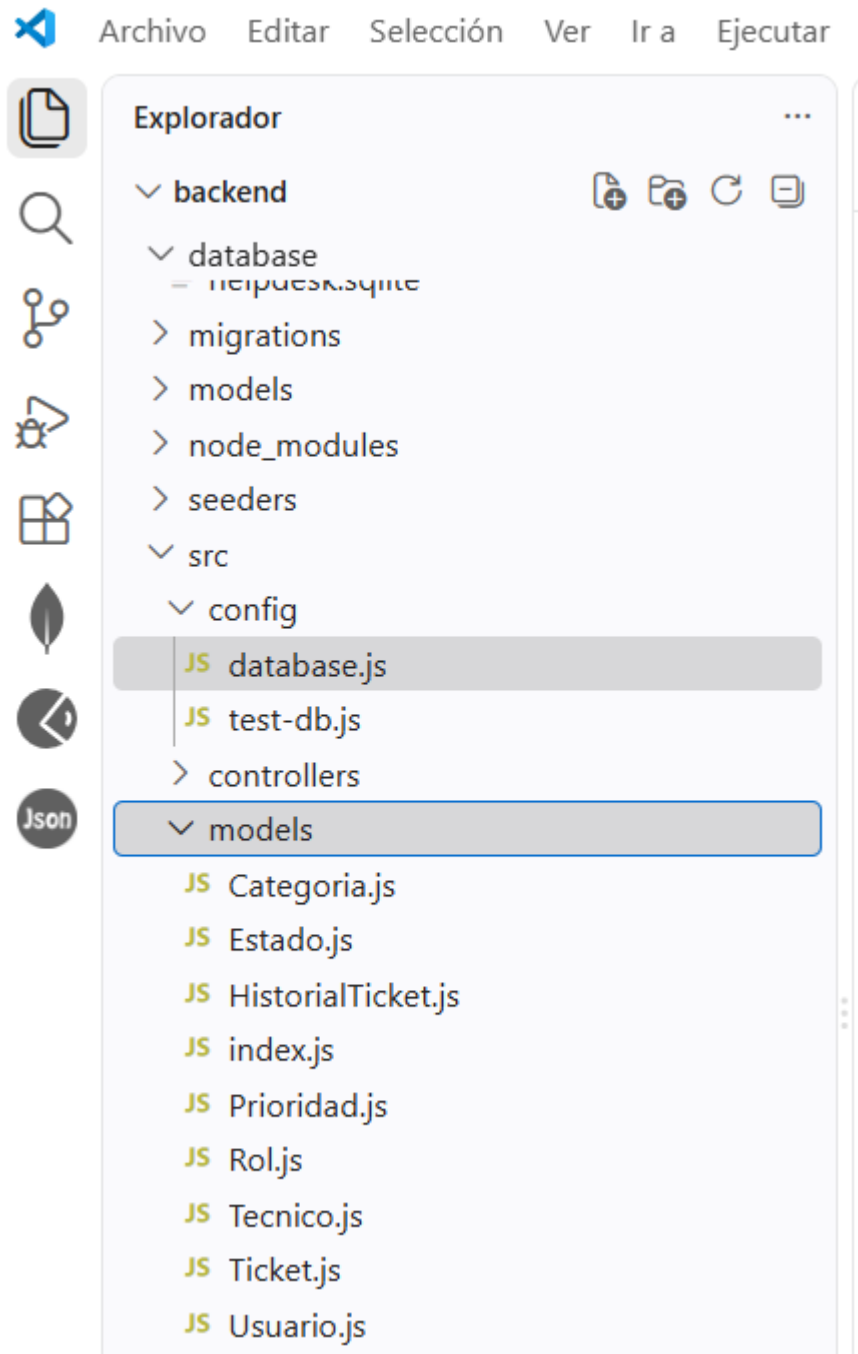


CAPTURA 3: Archivo database.js

7. MODELOS DEL SISTEMA

Modelos creados:

- Ticket
- Usuario
- Técnico
- Categoría
- Prioridad
- Estado
- HistorialTicket



CAPTURA 4: Carpeta models

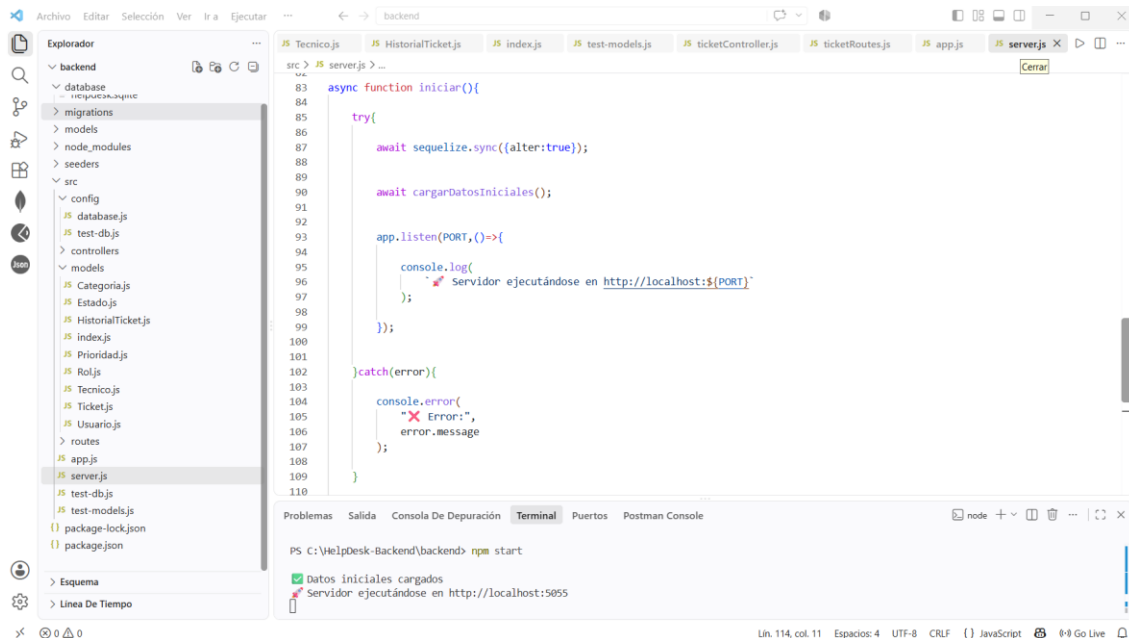
8. SERVIDOR EXPRESS

Puerto utilizado: 5055

Archivo server.js:

```
const app=require('./app');
```

```
const sequelize=require('./config/database');  
const PORT=5055;  
  
sequelize.authenticate();  
app.listen(PORT);
```



CAPTURA 5: Servidor ejecutándose

9. ENDPOINTS CRUD

GET /tickets - Listar tickets

GET /tickets/:id - Buscar ticket

POST /tickets - Crear ticket

PUT /tickets/:id - Actualizar ticket

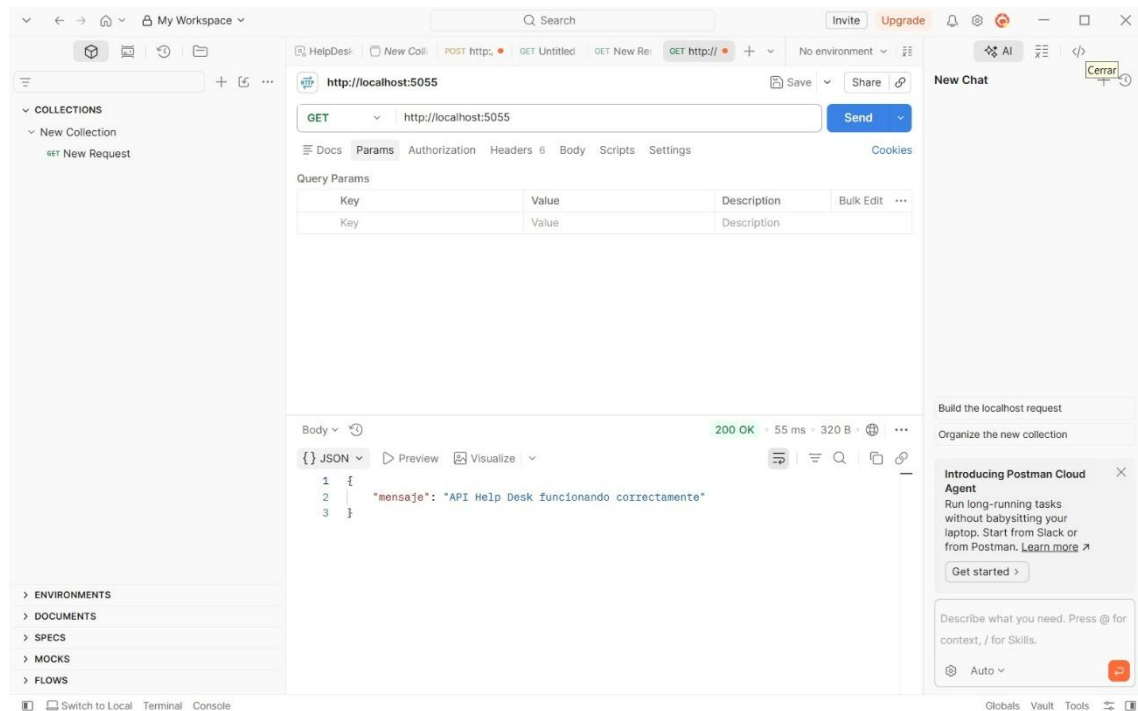
DELETE /tickets/:id - Eliminar ticket



10. PRUEBAS POSTMAN

01 - Verificar conexión API Help Desk

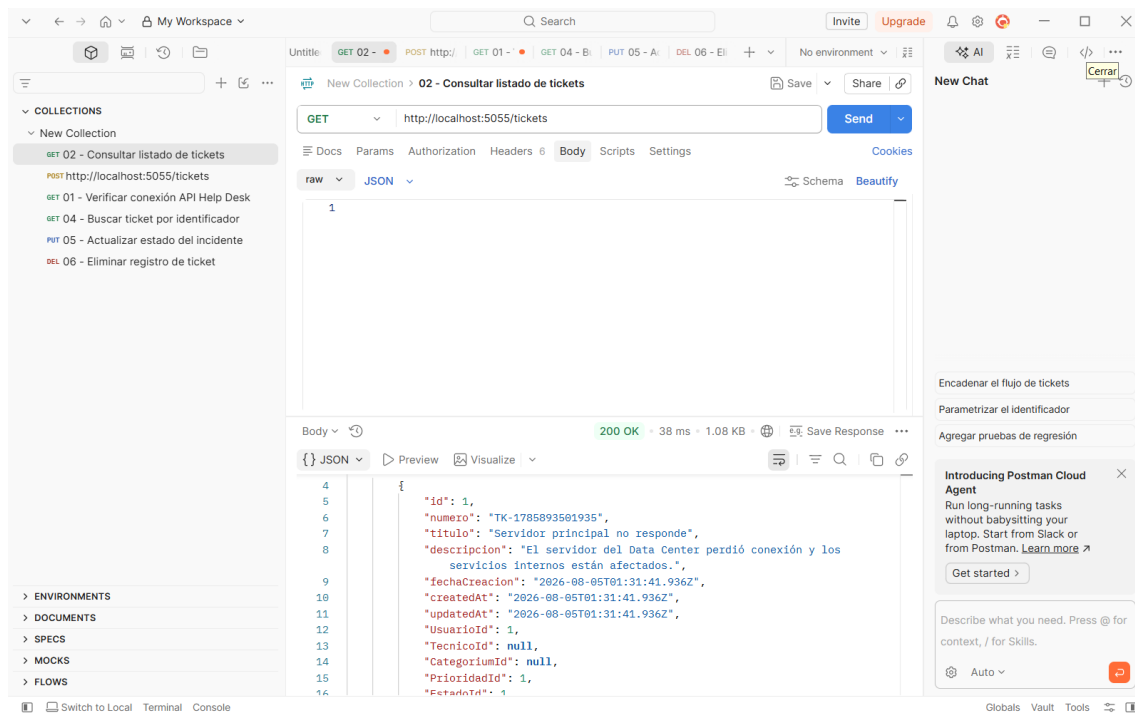
GET <http://localhost:5055/>



CAPTURA 6

02 - Consultar listado de tickets

GET <http://localhost:5055/tickets>



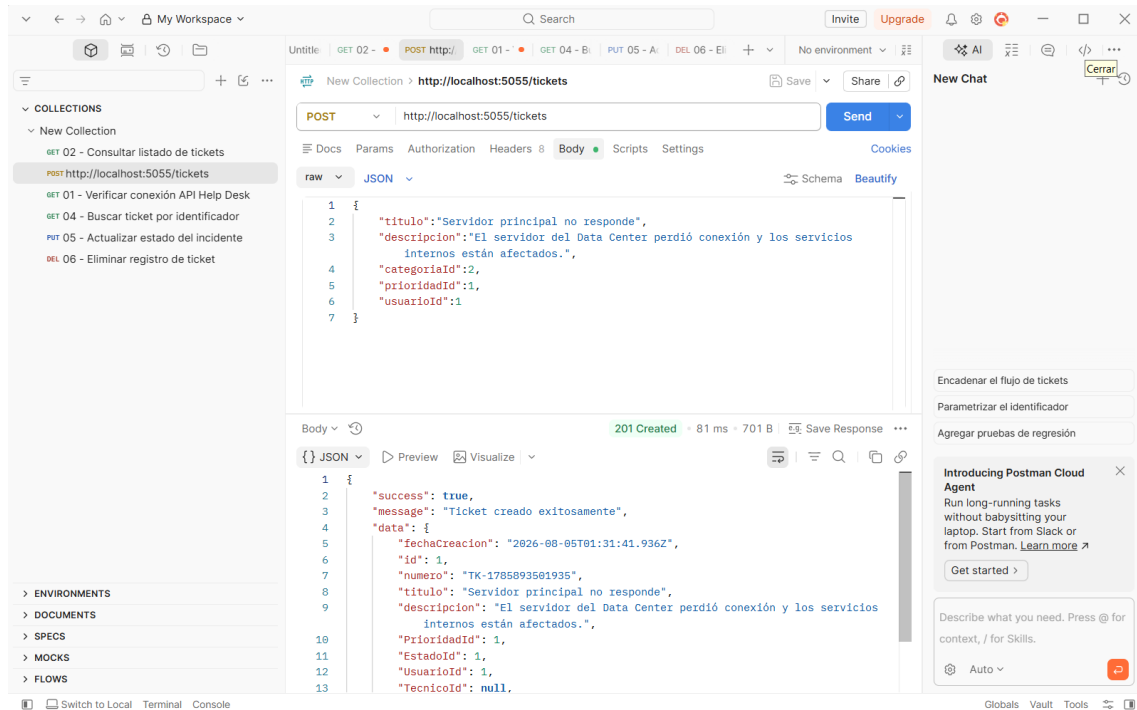
CAPTURA 7

03 - Registrar nuevo incidente

POST http://localhost:5055/tickets

Body:

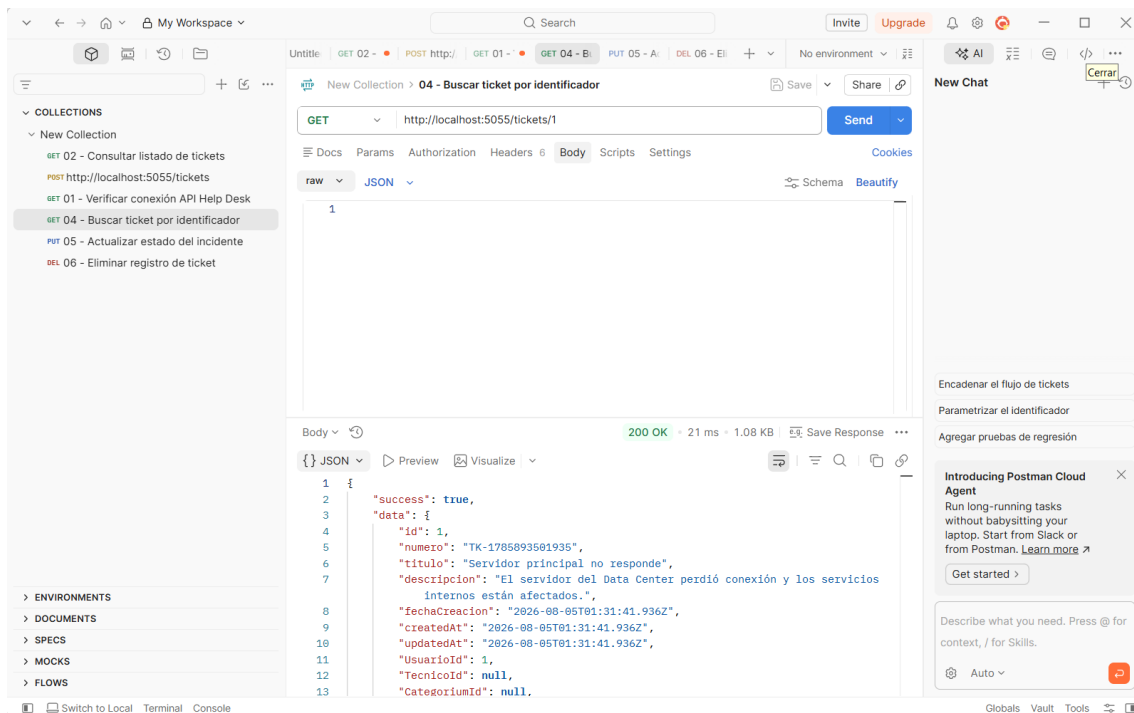
```
{
  titulo:'Servidor principal no responde',
  descripcion:'El servidor perdió conexión',
  categoriaId:2,
  prioridadId:1,
  usuarioId:1
}
```



CAPTURA 8

04 - Buscar ticket por ID

GET `http://localhost:5055/tickets/1`

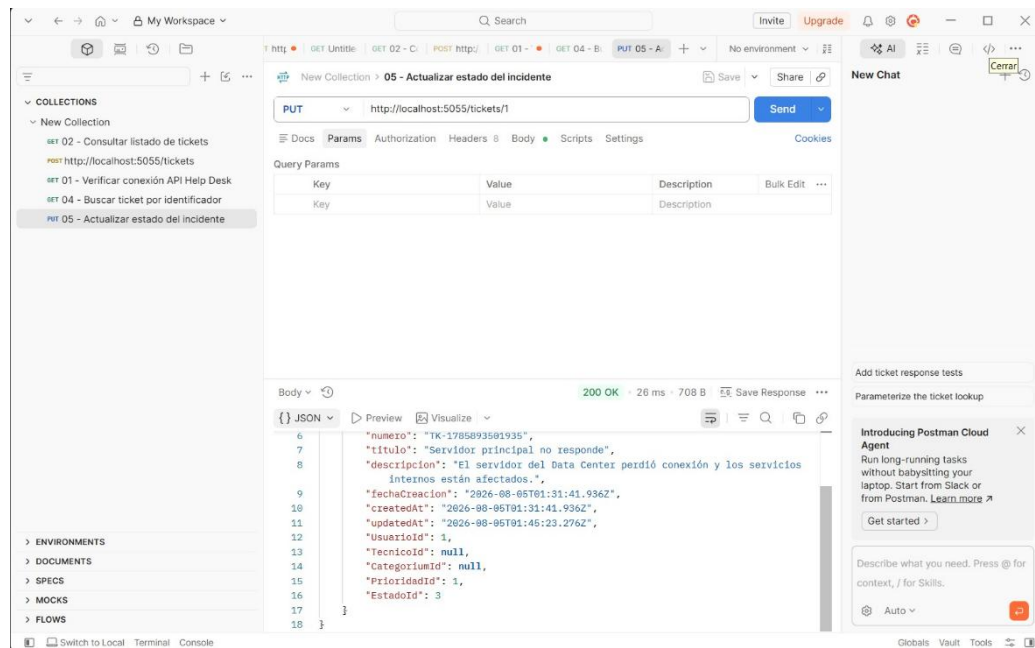


CAPTURA 9



05 - Actualizar estado

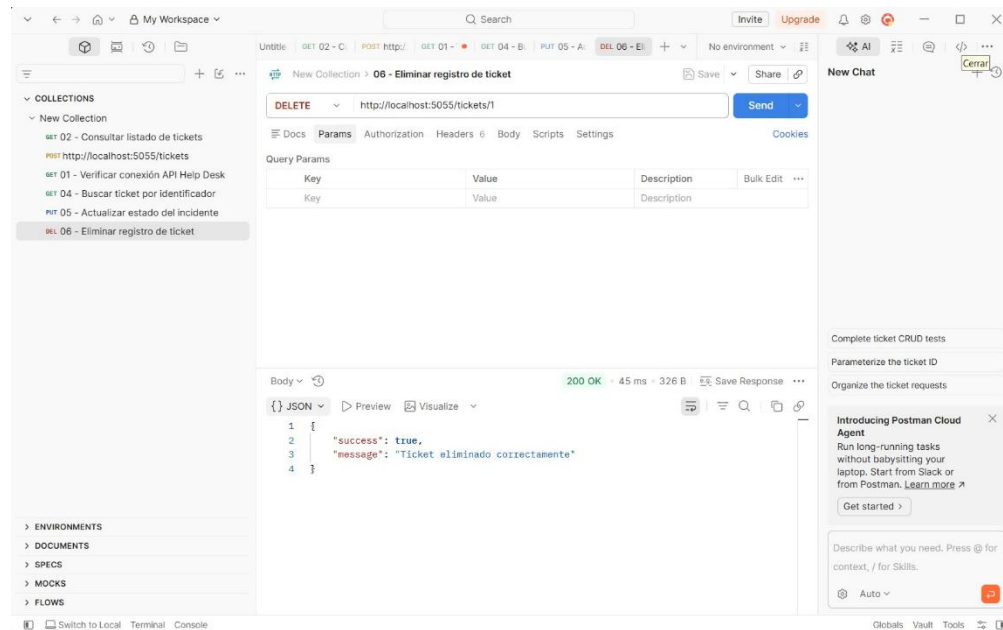
PUT <http://localhost:5055/tickets/1>



CAPTURA 10

06 - Eliminar ticket

DELETE <http://localhost:5055/tickets/1>



CAPTURA 11



11. CONTROL DE VERSIONES GIT

Rama utilizada:

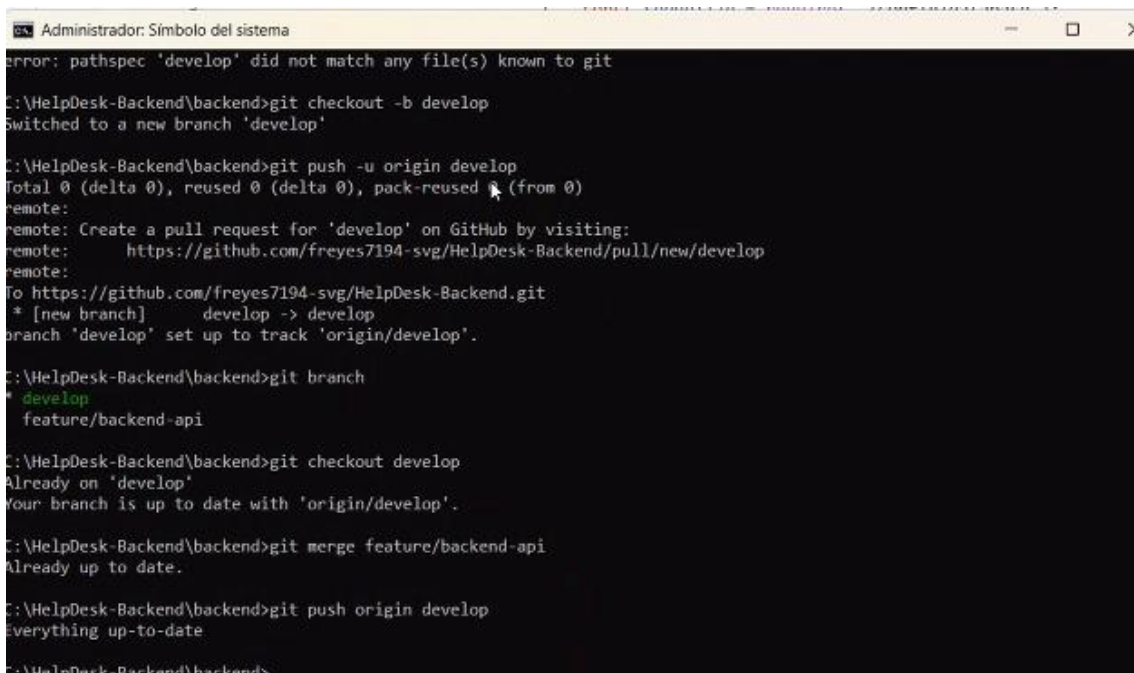
feature/backend-api

Comandos:

```
git checkout -b feature/backend-api
```

```
git add .
```

```
git commit -m 'Implementacion backend API Help Desk'
```



```
error: pathspec 'develop' did not match any file(s) known to git

C:\HelpDesk-Backend\backend>git checkout -b develop
Switched to a new branch 'develop'

C:\HelpDesk-Backend\backend>git push -u origin develop
Total 0 (delta 0), reused 0 (delta 0), pack-reused 1 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:   https://github.com/freyes7194-svg/HelpDesk-Backend/pull/new/develop
remote:
To https://github.com/freyes7194-svg/HelpDesk-Backend.git
 * [new branch]      develop -> develop
branch 'develop' set up to track 'origin/develop'.

C:\HelpDesk-Backend\backend>git branch
* develop
  feature/backend-api

C:\HelpDesk-Backend\backend>git checkout develop
Already on 'develop'
Your branch is up to date with 'origin/develop'.

C:\HelpDesk-Backend\backend>git merge feature/backend-api
Already up to date.

C:\HelpDesk-Backend\backend>git push origin develop
Everything up-to-date

C:\HelpDesk-Backend\backend>
```

CAPTURA 12: Evidencia GitHub

12. CONCLUSIONES

Se implementó el backend del Sistema de Gestión de Incidentes Help Desk utilizando Node.js, Express, Sequelize y SQLite. Las pruebas realizadas permitieron verificar el funcionamiento de los endpoints CRUD.



Evidencias

<https://github.com/freyes7194-svg/HelpDesk-Backend.git>

Bibliografía

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Silberschatz, A., Korth, H. F., & Sudarshan, S. (2019). *Database System Concepts* (7.^a ed.). McGraw-Hill Education.

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